

BHASKAR DAS

Montreal, QC, Canada (Permanent Resident) | Ph: +1 263-999-8098 | Email: bhaskardasrj@gmail.com
GitHub: github.com/das-bhaskar | LinkedIn: linkedin.com/in/bhaskardasrj

EDUCATION

Concordia University
Bachelor of Engineering, Software Engineering

Expected May 2027
Montreal, QC

EXPERIENCE

SamSearch — Software Engineer

Mar 2026 – Present

Vertical B2B SaaS platform for Government contracting

- Cut the AI activity log's query time from **8.45 seconds to 1.3 milliseconds**. A case-stripped filter was forcing a scan of 4 million rows; a planner-visible predicate let **PostgreSQL** use a 9,000-row partial index, built online.
- Built **CRM sync in both directions**: a deal created or moved on the pipeline board reaches the customer's CRM, and edits made there come back. Three connectors run on one provider-generic **OAuth** layer, which later carried mail, calendar and cloud storage for **Google and Microsoft**.
- Built the product's **reporting side**: one query engine over **20+ datasets**, a drag-and-drop dashboard builder, and click-through from a chart to the records behind it. Teams answer win-rate, trend and workload questions in the app instead of exporting to a spreadsheet.
- Brought the product's **LLM assistant** into **Slack and Teams**, running on the same agent as the web app. A team asks about their deals without leaving the chat, and anything that would change a record waits for a **person to tap approve**.
- Built the document pipeline behind **bid responses**, the most expensive manual step in a customer's week: it classifies each document in an incoming package, flags what is missing, and extracts the requirement set to check the draft response against.
- Migrated live records from single-user ownership to **organization ownership**, adding an owner column, an activity log, mentions and multi-assignee tasks. Teams who had bought seats could finally see and assign the same work.
- Gave customers running several brands a **separate workspace for each** inside one billing account, with its own documents, saved searches and assistant memory. Vector embeddings are tagged per workspace, so **no document surfaces in another brand's search**.
- Built behavioral usage tracking across the product. Server-side events write inside a **savepoint** on the request's own database session, so they are atomic with the mutation and **can never fail the request**; client events queue onto a **Redis Stream** drained separately.

Reliance Jio Ltd. — Software Engineering Intern, Speech Systems & AI

May 2024 – Jul 2024

- Built a **speaker diarization** pipeline that labels who is speaking in noisy webcam audio, segmenting the stream and clustering **NVIDIA TitaNet** voice embeddings by cosine similarity into **per-speaker captions**.
- Took **Whisper** transcription to high confidence on inconsistent recordings by resampling to **16 kHz** and applying 7x gain only to segments a **voice activity detector** marked as speech.

PROJECTS

AetherShell — C, Swift (ARKit), Java (Spring Boot), Python

github.com/das-bhaskar/AetherShell

- **Featured on Hackaday** and released on the **App Store**: an open-source bridge that streams sketches drawn on a **Nintendo DSi** into an iOS app, which anchors them as **AR animations** to real surfaces via ARKit and a Python vision engine.
- Kept strokes continuous over unstable legacy **Wi-Fi** by streaming stylus coordinates off the DSi over **UDP** and filtering out points thrown by ARM9 lag, and got the handheld past security landing pages with custom HTTP headers and **SSL termination** on a Java hub.

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, C++, Swift, SQL, Bash

Backend & Data: FastAPI, Spring Boot, PostgreSQL, Redis Streams, REST, OAuth 2.0, WebSockets, Docker

Frontend: React, ARKit

AI: LLM agent tooling, RAG, vector search, Whisper, TitaNet, VAD, CUDA, Gemini API